

EQUIPMENT SUBMITTAL

Clean Agent System

Project: Meridian Data Centre Campus - Phase 1

Reference Section: 21 22 00

Package ID: SUB-212200-01-R0

Revision: R0

Reviewed Spec Revision: Rev 2024

Date: 01-01-2025

ACTIVE

Rev	Date	Description
R0	01-01-2025	Initial / Updated Submittal

Technical Parameters

Parameter	Claimed Value
Suppressant Agent Type	FK-5-1-12
Total Clean Agent Agent Quantity	11800 kg
Cylinder Operating Pressure	360 psi
Design Concentration	4.7%
NOAEL Safety Limit	10%
Standards Reference	NFPA 2001-2008

¹ discharge time extends to 11.0 seconds when piping runs exceed 45 meters

² Discharge Time is 9.2 s (standard pipe configuration)

Clause ID	Requirement Summary	Response
21 22 00 Part 1.3.3	Factory test and compliance certificates issued by UL and FM for all clean agent storage cylinder va...	Comply
21 22 00 Part 1.2.1	The clean agent fire suppression system shall be designed, installed, and tested in accordance with ...	NFPA 2001 (2008 Edition)
21 22 00 Part 2.2.3.A	Clean agent discharge time shall not exceed 10 seconds from the moment the discharge valve opens unt...	Comply
21 22 00 Part 2.2.3.B	Agent hold time (retention) shall be a minimum of 10 minutes (600 seconds) to prevent re-ignition. H...	9 minutes
21 22 00 Part 2.3.4.F	The vendor shall submit a NOAEL safety margin calculation demonstrating that the actual maximum agen...	Comply
21 22 00 Part 2.3.1.A	Clean agent storage cylinders shall be pressurised to 500 psi (34.5 bar) with nitrogen super-pressur...	Comply
21 22 00 Part 2.2.5	Minimum total clean agent quantity for the campus shall be 12,500 kg of FK-5-1-12. Agent quantity pe...	Comply
21 22 00 Part 1.2.3	All clean agent storage cylinders and discharge valve assemblies shall be tested, listed, and labele...	ISO 9809-1
21 22 00 Part 2.1.1	The clean agent shall be FK-5-1-12 (Dodecafluoro-2-methylpentan-3-one), chemically identified as CF3...	Comply
21 22 00 Part 2.2.1	Design concentration shall be 4.7% by volume (v/v) for Class A and Class C fire hazards. The design ...	Comply

Clause ID	Requirement Summary	Response
21 22 00 Part 2.2.2	The No Observed Adverse Effect Level (NOAEL) for FK-5-1-12 is 10.0% by volume (v/v). The design conc...	Comply